Who should get a retention offer?

Profit-aware churn and retention targeting - Business Analytics capstone, Topic 7

- Decision: which customers receive an offer, with which offer, and how much budget to commit
- Data: IBM Telco Customer Churn sample (7,043 customers)
- Recommendation: expected-value targeting, tested in a randomised pilot

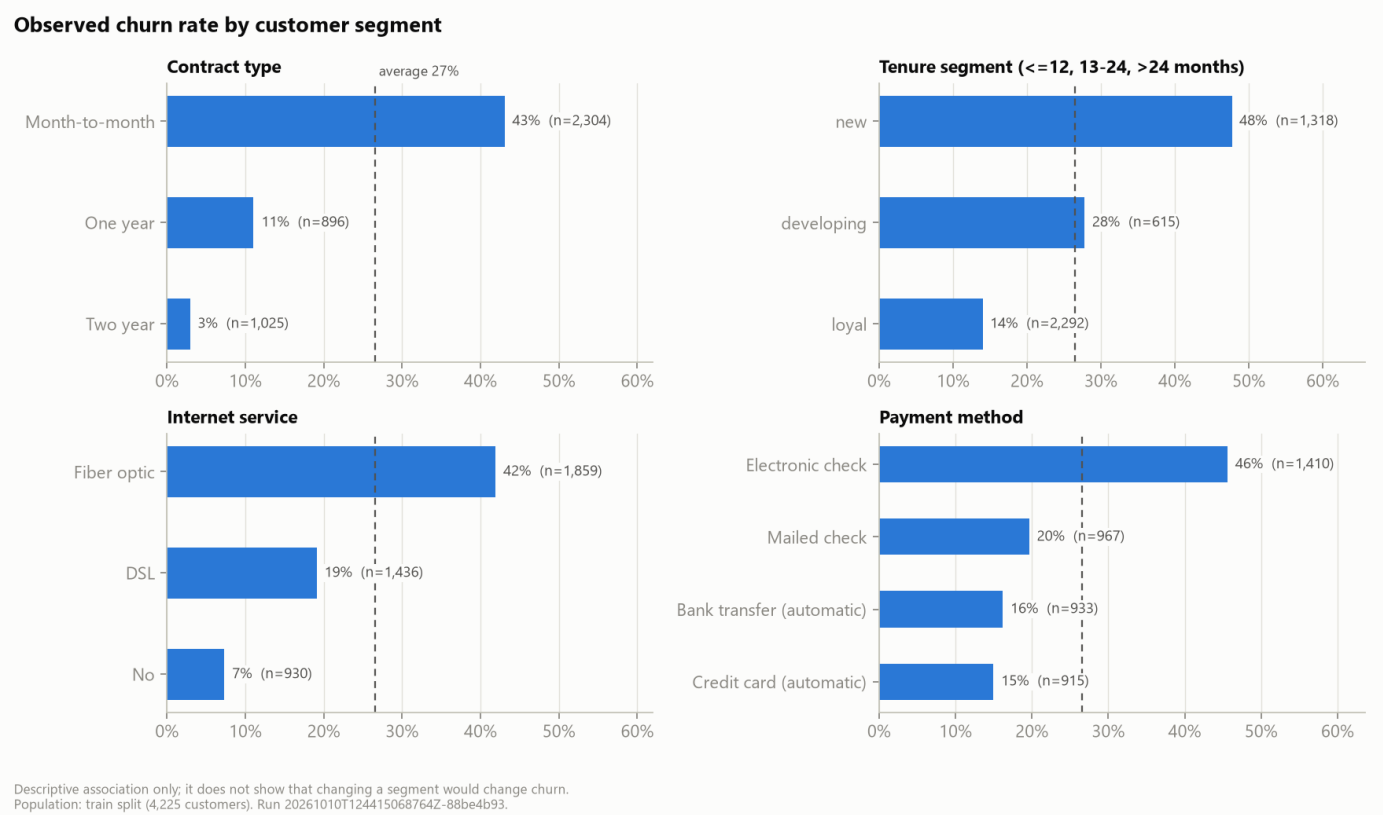
Today's rule wastes part of the budget

Calling the riskiest customers ignores what they are worth and what the offer costs

- Budget covers an offer for only 10% of customers ($3,500 per 1,409)
- Every offer costs money even if the customer would have stayed
- A very risky customer with a small bill may not repay the offer
- Decision owner: Head of Retention; Finance approves budget; Legal/DPO approves contact

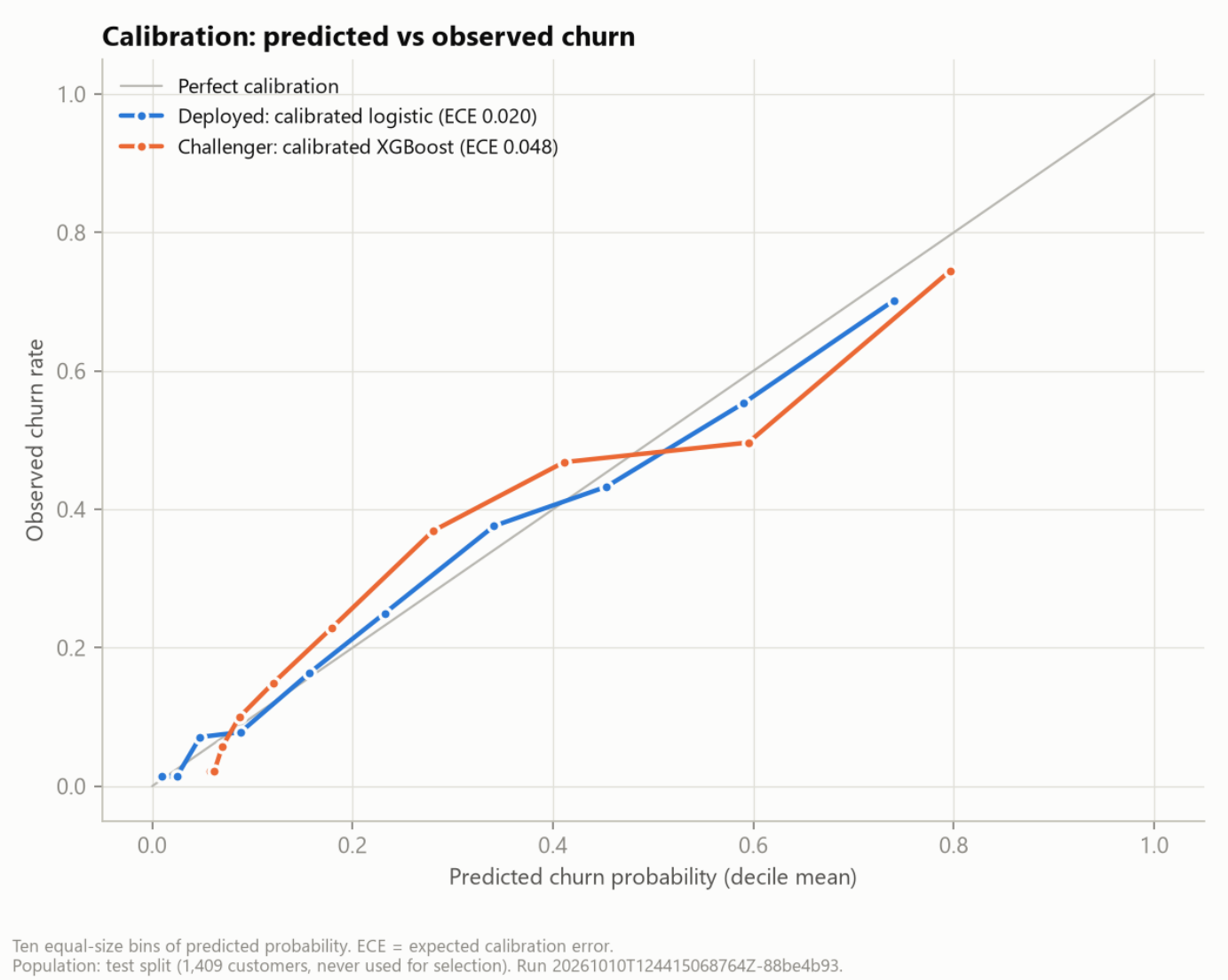
The data: clean, but a teaching sample

- 26.5% churn; month-to-month, new, fibre and e-cheque customers churn most
- No offer or campaign data: save rates must be assumed
- Fictional US company; real US postpaid churn is about 1% a month



A calibrated model ranks churners well

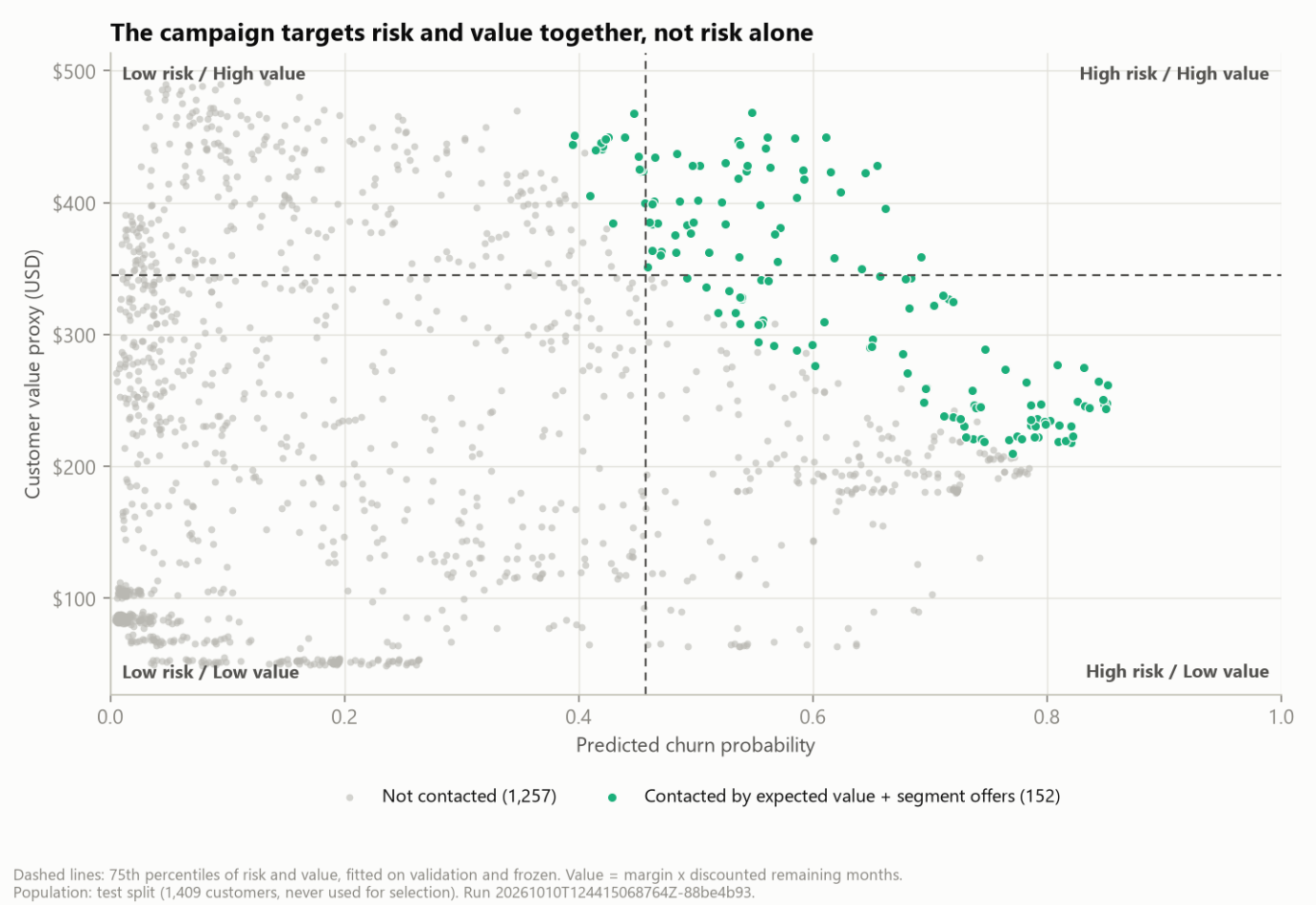
- Logistic regression chosen over random forest and XGBoost by a pre-set rule
- PR-AUC 0.62 vs 0.27 random; ROC-AUC 0.83
- Probabilities off by 2.0% on average (ECE)
- Predicts who leaves, not who an offer keeps



Ten equal-size bins of predicted probability. ECE = expected calibration error.
Population: test split (1,409 customers, never used for selection). Run 20261010T124415068764Z-88be4b93.

Risk and value only partly overlap

- Risk-only spends almost the whole budget on high-risk, low-value customers
- The recommended rule reaches all 51 high-risk, high-value customers
- It adds valuable customers at moderate risk



The decision rule

Expected value = p(churn) x save rate x value - (offer + contact cost)

- Contact only customers with positive expected value
- Optimiser picks customer and offer within budget, one offer each, max 30% contacted
- Four offers with eligibility rules: credit, contract upgrade, service bundle, loyalty reward
- All costs and save rates are visible, sourced assumptions

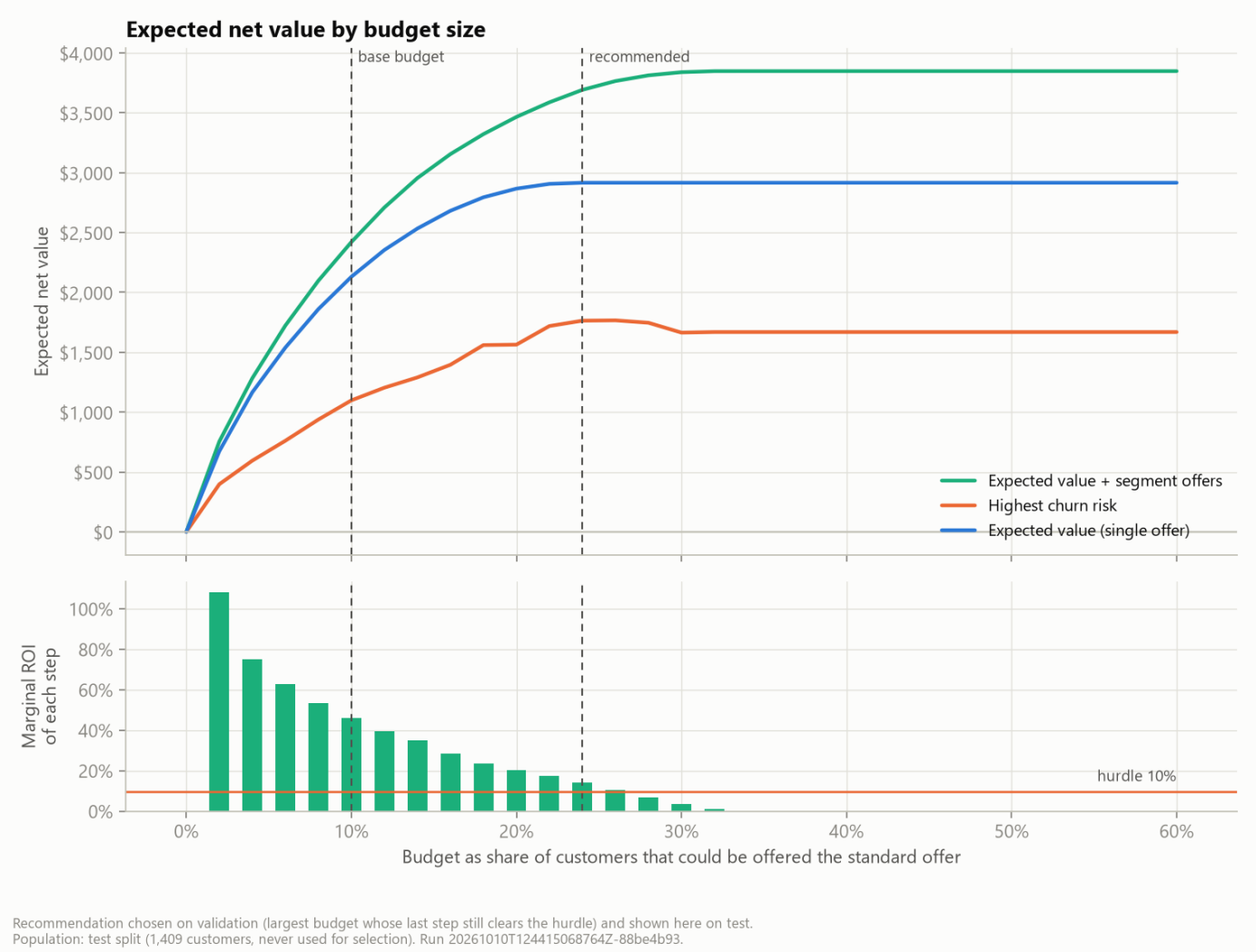
Same budget, more value

- Better targeting: +$1,032; with actual outcomes +$1,308 (95% CI $509 to $2,153)

Rule	Customers	Net value	ROI	HR/HV coverage
Highest churn risk	140	$1,098	31%	4%
Expected value	140	$2,130	61%	96%
EV + segment offers	152	$2,420	69%	100%

How much to spend

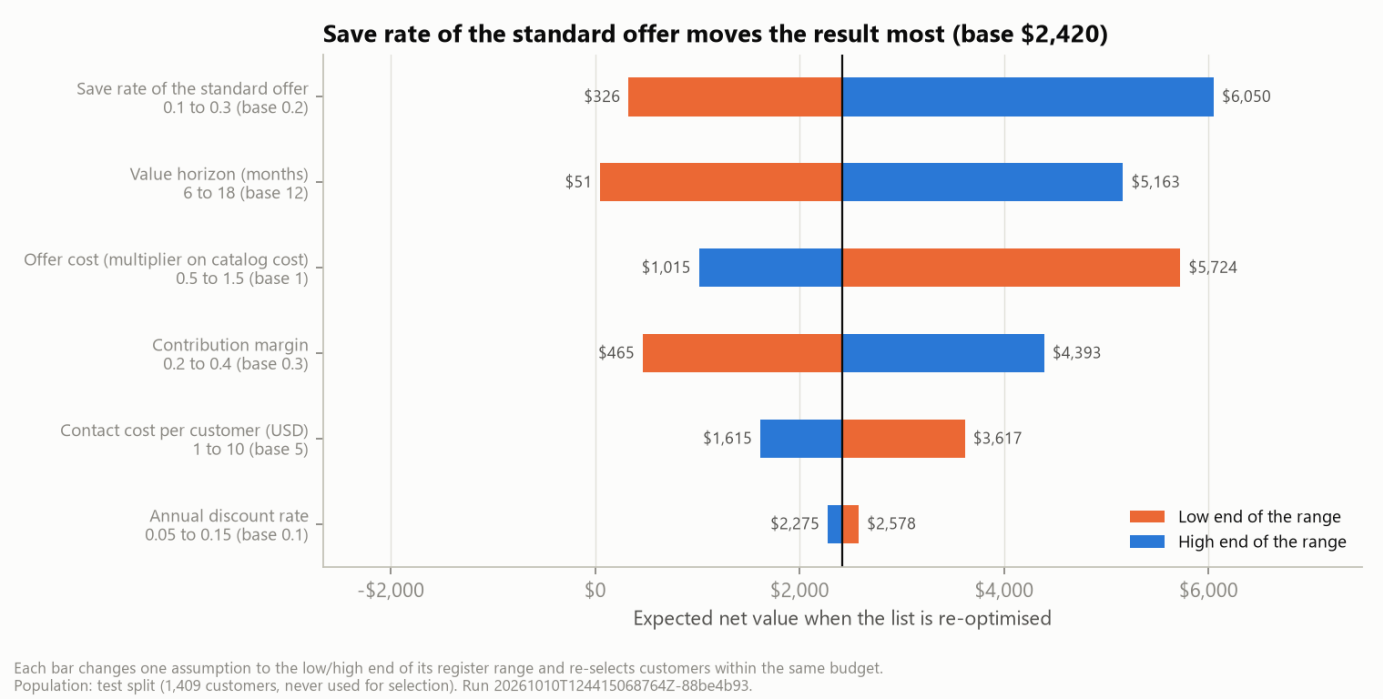
- Recommended: 24% of customers (chosen on validation)
- Test: 310 offers, $8,450, net $3,692 (ROI 44%)
- Stop adding budget when the next step earns less than 10%



Recommendation chosen on validation (largest budget whose last step still clears the hurdle) and shown here on test.
Population: test split (1,409 customers, never used for selection). Run 20261010T124415068764Z-88be4b93.

How robust is it?

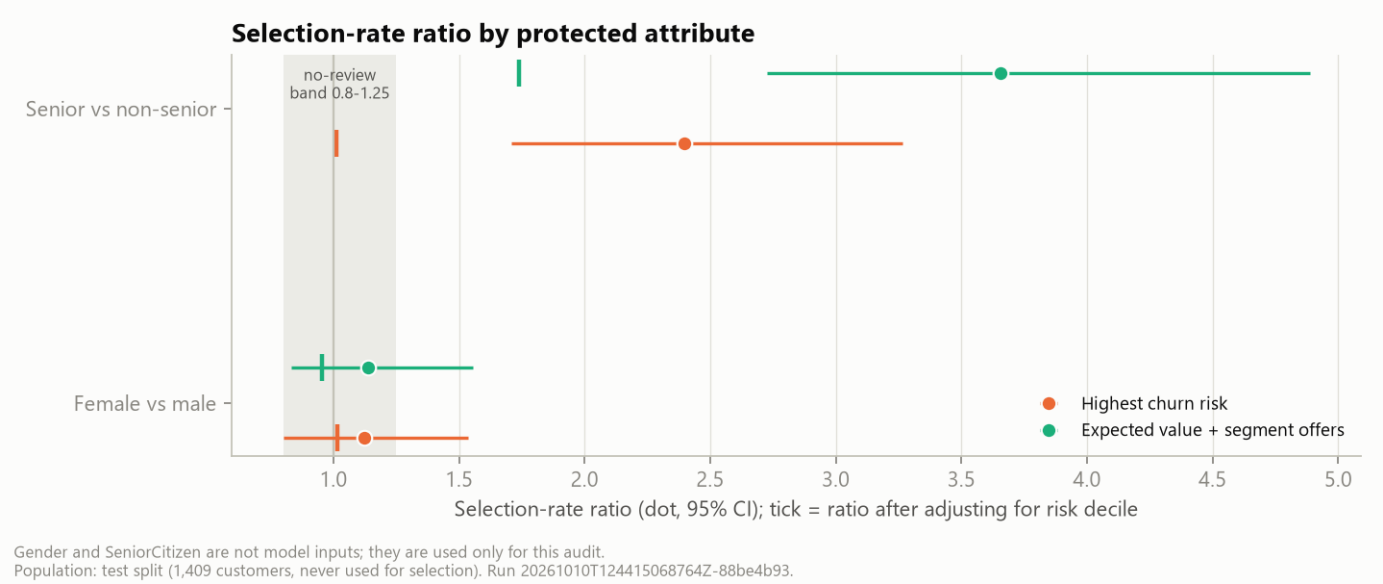
- Break-even save rate 11.8% (assumed 20%)
- Pessimistic $0 (no campaign) / base $2,420 / optimistic $18,525
- Below 8.9% save rate: run no campaign



Each bar changes one assumption to the low/high end of its register range and re-selects customers within the same budget.
Population: test split (1,409 customers, never used for selection). Run 20261010T124415068764Z-88be4b93.

Fairness and customer experience

- Seniors contacted 3.7x as often - mostly higher risk, partly higher value
- No protected attribute is a model input; every wave is audited
- Loyal low-risk customers rarely get offers: add a loyalty benefit



Gender and SeniorCitizen are not model inputs; they are used only for this audit.
Population: test split (1,409 customers, never used for selection). Run 20261010T124415068764Z-88be4b93.

Risks and limits

- No causal evidence: save rates are assumptions; contact can even raise churn
- Teaching sample: fictional US data, 26.5% churn vs about 1% a month in reality
- Vietnam: about 90% prepaid; rebuild the model on our own data
- Compliance: consent, Do-Not-Call, calls 08:00-17:00, one call per 24 hours, DPIA

The pilot measures what we assumed

- One wave: 1,690 customers, 845 offered, 845 held out at random
- Outcome: churn after 90 days; save rate = 1 - churn(offered) / churn(holdout)
- Detects a save rate of about 11% or more in one wave
- Offer and contact cost about $23,033

Decisions requested

- Approve one pilot wave sized at 24% of the base, half held out
- Name owners: Retention, Call centre, Finance, Legal/DPO
- Agree the go rule now: scale only if the save rate's lower 95% bound is above 11.8%
- Live demo: dashboard - KPIs, action list, scenarios